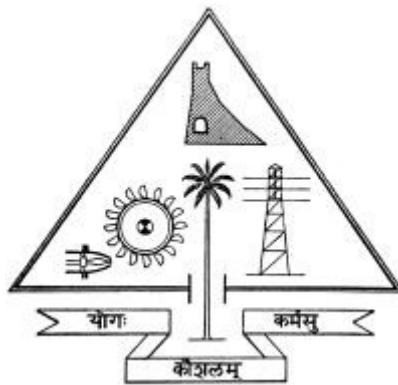


ADAPTIVE MULTI-AGENT DEFENSE SYSTEM AGAINST ADVERSARIAL ATTACKS ON LLMS

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ABSTRACT

Large Language Models (LLMs) are widely used in modern software systems for applications such as conversational agents, automated content generation, and enterprise decision support. However, they remain vulnerable to adversarial prompt-based attacks, including jailbreaks and prompt injections, which can bypass existing safety mechanisms. Current defense approaches struggle to adapt to evolving attack strategies, creating significant security risks and highlighting the need for adaptive and robust protection mechanisms.

This work proposes an Adaptive Multi-Agent Defense Framework that combines evolutionary attack generation, multi-layer detection, and real-time adaptation to secure LLM-based systems. The framework comprises an evolutionary attack generation engine, a multi-agent detection system with specialized verifier agents, and an adaptive defense architecture that updates strategies based on observed attack patterns. Its defense pipeline includes input sanitization, anomaly detection, multi-verifier consensus, and output validation, enabling comprehensive protection across syntactic, semantic, behavioral, and contextual attack vectors.

A key contribution of the proposed system is the use of semantic-aware evolutionary algorithms to generate realistic adversarial prompts for proactive stress-testing and robust training data generation. Detection decisions are produced through a probabilistic consensus mechanism that balances detection accuracy with false-positive reduction, maintaining usability for legitimate queries. The framework also integrates online learning and explainable decision components, enabling continuous adaptation and transparent security decisions within a unified, deployable defense solution.

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